

Solvency Cones and Super-hedging

**IN MARKETS WITH
TRANSACTION COSTS**

Presented By - Aryan Chaturvedi & Tanay Pathode



Problem Statement

Title: Superhedging in Markets with Transaction Costs

- Given:**
- A discrete-time financial market with
 $t = 0, 1, \dots, T$
 - Multiple assets (d assets)
 - Transaction costs (bid-ask matrix π_{ij})

The Goal: Find all initial portfolios that can superhedge a contingent claim

Mathematical Tools

Transaction Matrix

$$\Pi_t = \begin{matrix} \downarrow \\ \omega \rightarrow \Pi_t(\omega) \end{matrix} \begin{matrix} \text{Grid of } (d \times d) \text{ cells} \end{matrix}$$



Encodes transaction costs + exchange rates



$\pi^{ij} \rightarrow$ Number of “Asset i” needed to buy 1 unit of “asset j”



$(\frac{1}{\pi^{ji}}, \pi^{ij}) \rightarrow$ Bid and Ask-prices of “Asset j” in terms of “Asset i”



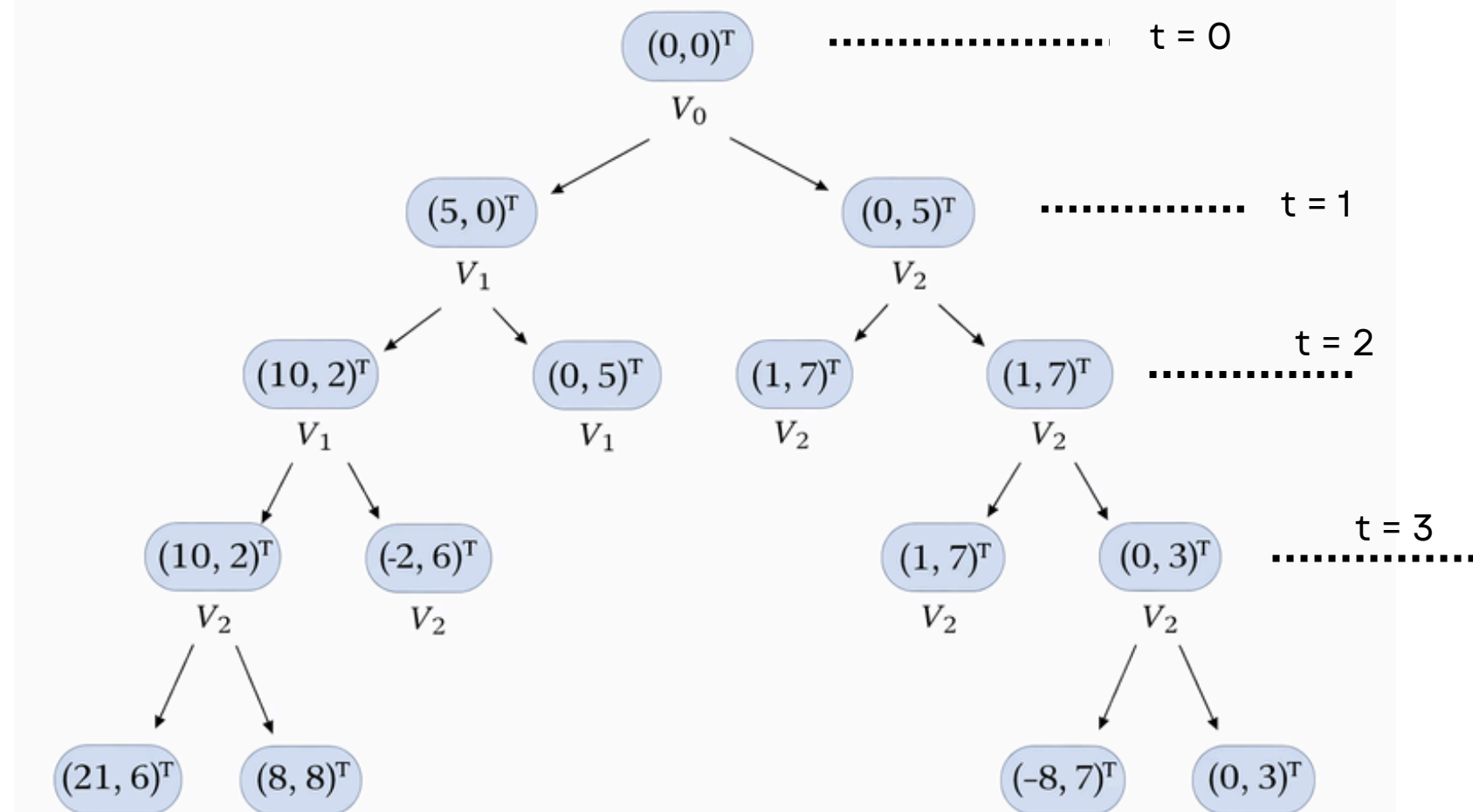
$\pi^{ij} > 0, \quad 1 \leq i, j \leq d, \rightarrow$ Prices are always positive



$\pi^{ij} \leq \pi^{ik} \cdot \pi^{kj} \rightarrow$ No arbitrage via indirect trading

Portfolio vector V_t :

- V_t is $\mathcal{F}_t$ -measurable
- A portfolio vector at time t $\rightarrow V_t : \Omega \rightarrow \mathbb{R}^d$
- For each state ω , $\rightarrow V_t(\omega) = (v_1, v_2, \dots, v_d)$
- $A_T : \{V_T\}$, or A_T is set of superhedgeable claims starting from initial endowment $0 \in \mathbb{R}^D$ at $t=0$.



Mathematical Tools

► Solvency Cones $\rightarrow K_t(\omega)$

- At time 't' and state ' ω ': $K_t(\omega) \subset \mathbb{R}^d$
- $K_t(\omega)$ is a polyhedral convex cone
- Trading Vectors $\rightarrow \{ \pi^{ij}(\omega) e_i - e_j, e_i \}$
- $K_t(\omega) = \text{cone} \{ \pi^{ij}(\omega) e_i - e_j, e_i \mid i, j = 1, \dots, d \}$

↪ It is negative of basic exchange trades.

- **Adaptedness:** K_t is $\mathcal{F}_t$ -measurable

- Solvency Property: If $x \in K_t$, then:

$$\exists v \in -K_t \text{ s.t. } (x + v) \in \mathbb{R}_+^d$$

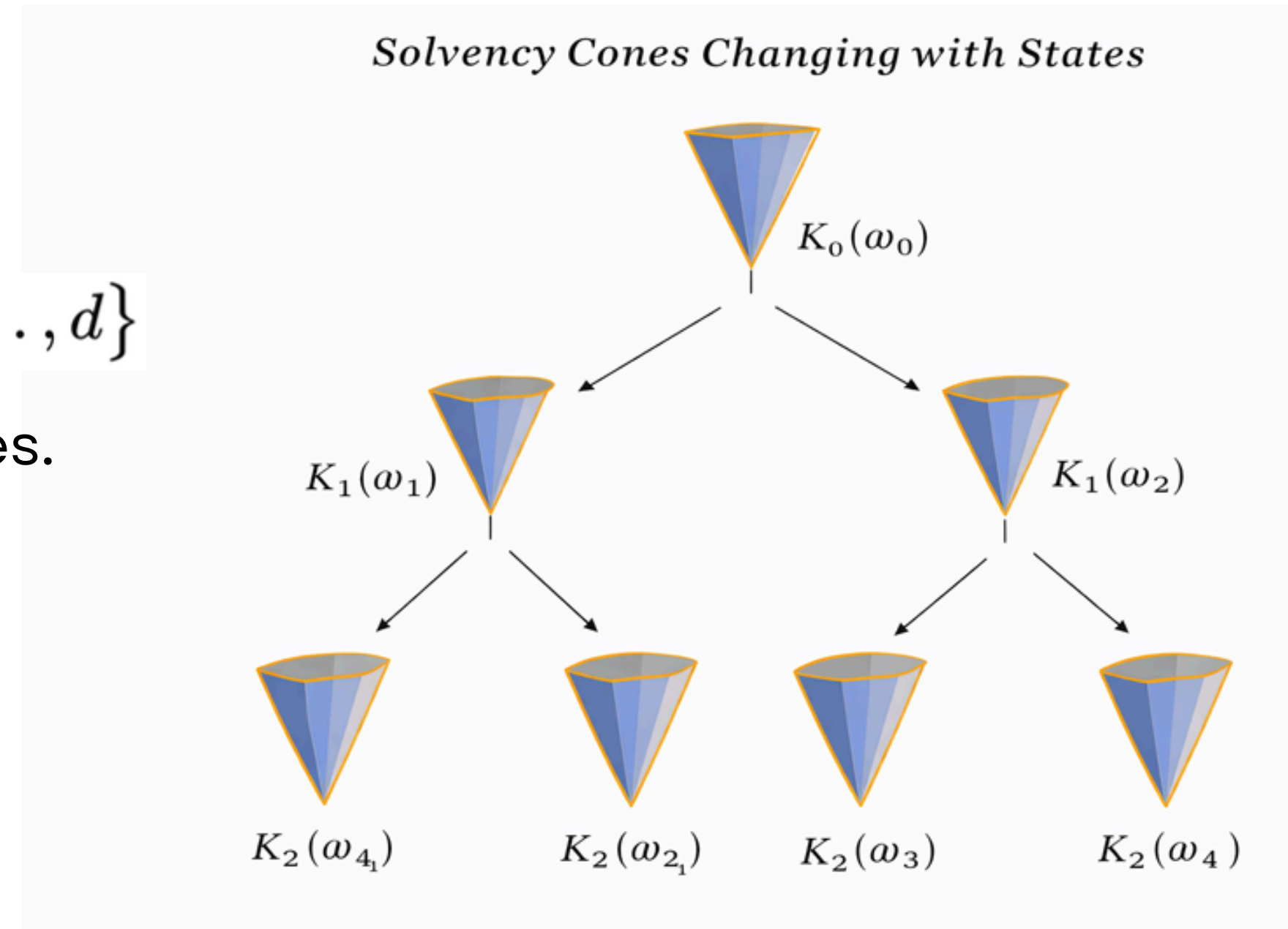
- Financial Meaning of Cones

➤➤ Set of All Feasible Trades

➤➤ Cone represents trade takes place in only one direction.

➤➤ Changes in a self financing portfolio can only be made via trades in mkt:

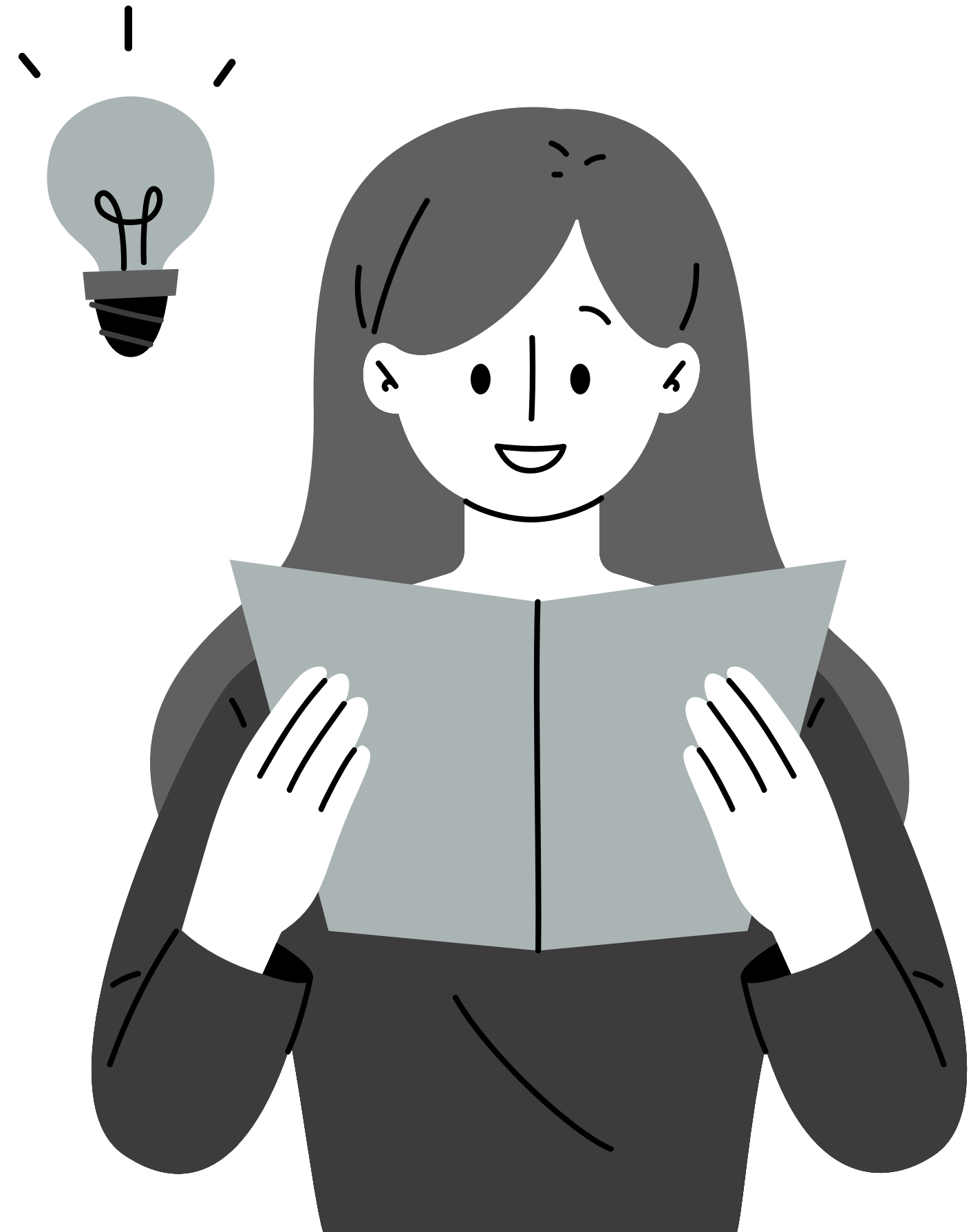
$$V_t - V_{t-1} \in -K_t$$



Mathematical Tools

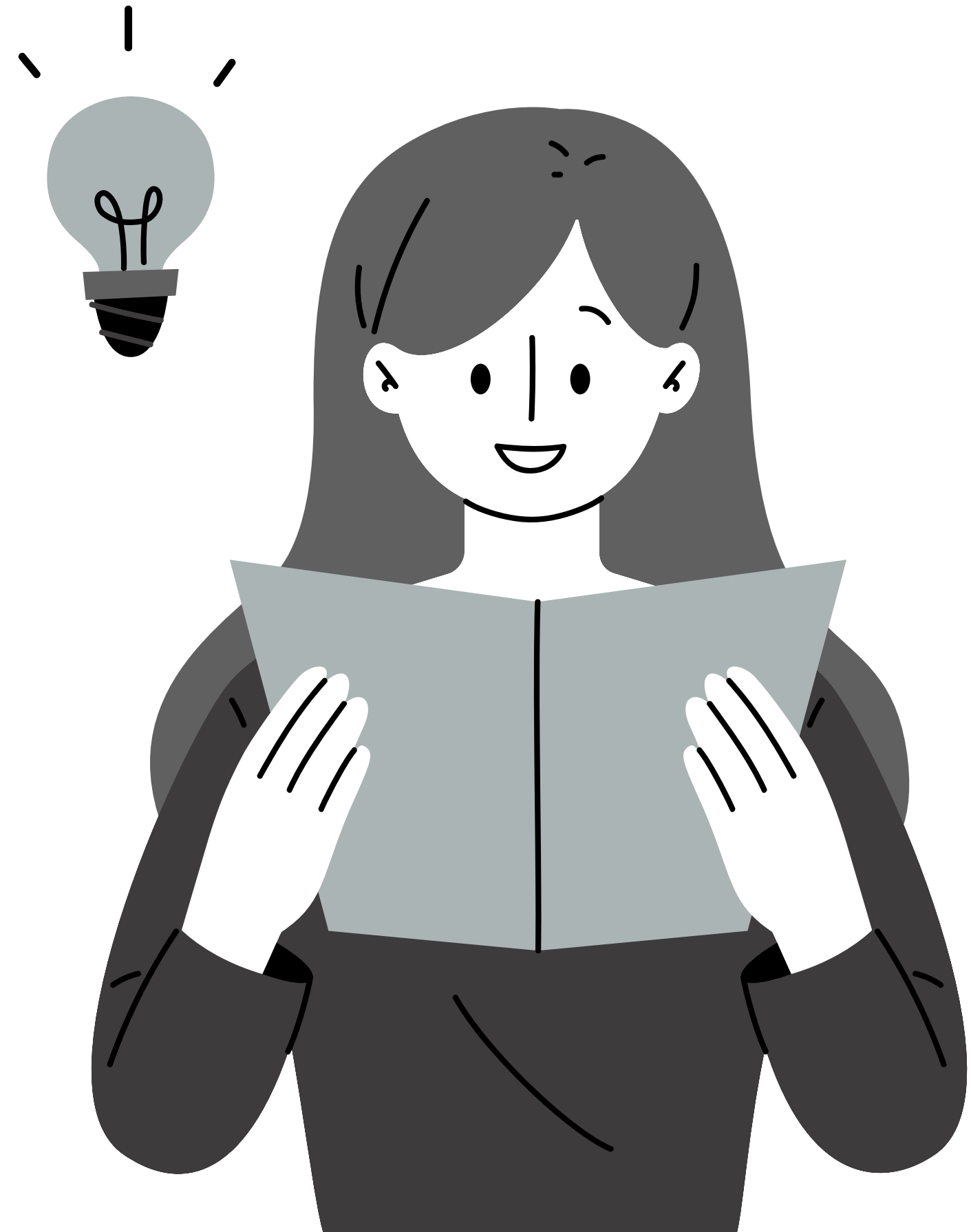
Market Model

- A market model is an $(\mathcal{F}_t)_{t=0}^T$ adapted process given by: $(K_t)_{t=0}^T$, where each K_t is a **solvency cone**
- The process (K_t) is **adapted to filtration** $(\mathcal{F}_t)$
- At each time t and state ω : $K_t(\omega) \subset \mathbb{R}^d$
- It represents all feasible trades under transaction costs
- Available Trades in the Market $\rightarrow$
 1. Exchange of one asset for another
 2. Disposal of assets (free disposal)
- $L_d^0(\mathcal{F}_t, \mathbb{R}^d) = \{ X : \Omega \rightarrow \mathbb{R}^d \mid X \text{ is } \mathcal{F}_t\text{-measurable} \}$
 - ↪ set of all $\mathcal{F}_t$ -measurable random vectors $\longleftrightarrow X : \Omega \rightarrow \mathbb{R}^d$
- $A_T \subseteq L_d^0(\mathcal{F}_T, \mathbb{R}^d) \longleftrightarrow$ the set of random vectors $\longrightarrow V_T : \Omega \rightarrow \mathbb{R}^d$
 - ↪ $A_T = -L_d^0(\mathcal{F}_0, K_0) - L_d^0(\mathcal{F}_1, K_1) - \dots - L_d^0(\mathcal{F}_T, K_T)$



Important Results

- ▶ No Arbitrage Condition $\longleftrightarrow A_T \cap L_d^0(\mathcal{F}_T, \mathbb{R}_+^d) = \{0\}$
- ▶ **Dual Cone** $K_t^+ \longleftrightarrow K_t^+(\omega) = \{v \in \mathbb{R}^d \mid v^T u \geq 0 \ \forall u \in K_t(\omega)\}$
 - ↪ Set of consistent price directions
Prices that respect transaction costs
- ▶ **Fundamental Theorem of Asset Pricing:**
 - No Arbitrage $\Leftrightarrow$ Exists Consistent Price System (Z_t)
 - **Properties of Z_t** $\begin{cases} Z_t \in K_t^+ \\ Z_t \text{ is a martingale} \end{cases}$
 - Instead of a single price a set of consistent price systems that respect transaction costs."
- ▶ A portfolio $x_0 \in \mathbb{R}^d$ **superhedges** X if: $X \in x_0 + A_T$
- ▶ Set of Superhedging Portfolios:
 - ↪ $SHP_0(X) = \{x_0 \in \mathbb{R}^d \mid X \in x_0 + A_T\}$



Formulations

DUAL REPRESENTATION

$$SHP_0(X) = \{x_0 \in \mathbb{R}^d \mid \forall Z \in \mathcal{Z} : \mathbb{E}[X^T Z_T] \leq x_0^T Z_0\}$$

$$SHP_0(X) = \bigcap_{(Q,w) \in W^1} (\mathbb{E}^Q[X] + G(w))$$

Where:

Half-space: $G(w) = \{x \in \mathbb{R}^d \mid w^T x \geq 0\}$

Dual Set W: $W^1 = \left\{ (Q, w) \in \mathcal{M}_{1,d}^P \times \mathbb{R}^d \setminus \{0\} \mid \forall t : \mathbb{E} \left[\text{diag}(w) \frac{dQ}{dP} \mid \mathcal{F}_t \right] \in K_t^+ \right\}$

Formulations

RECURSIVE CONSTRUCTION

Terminal Step:

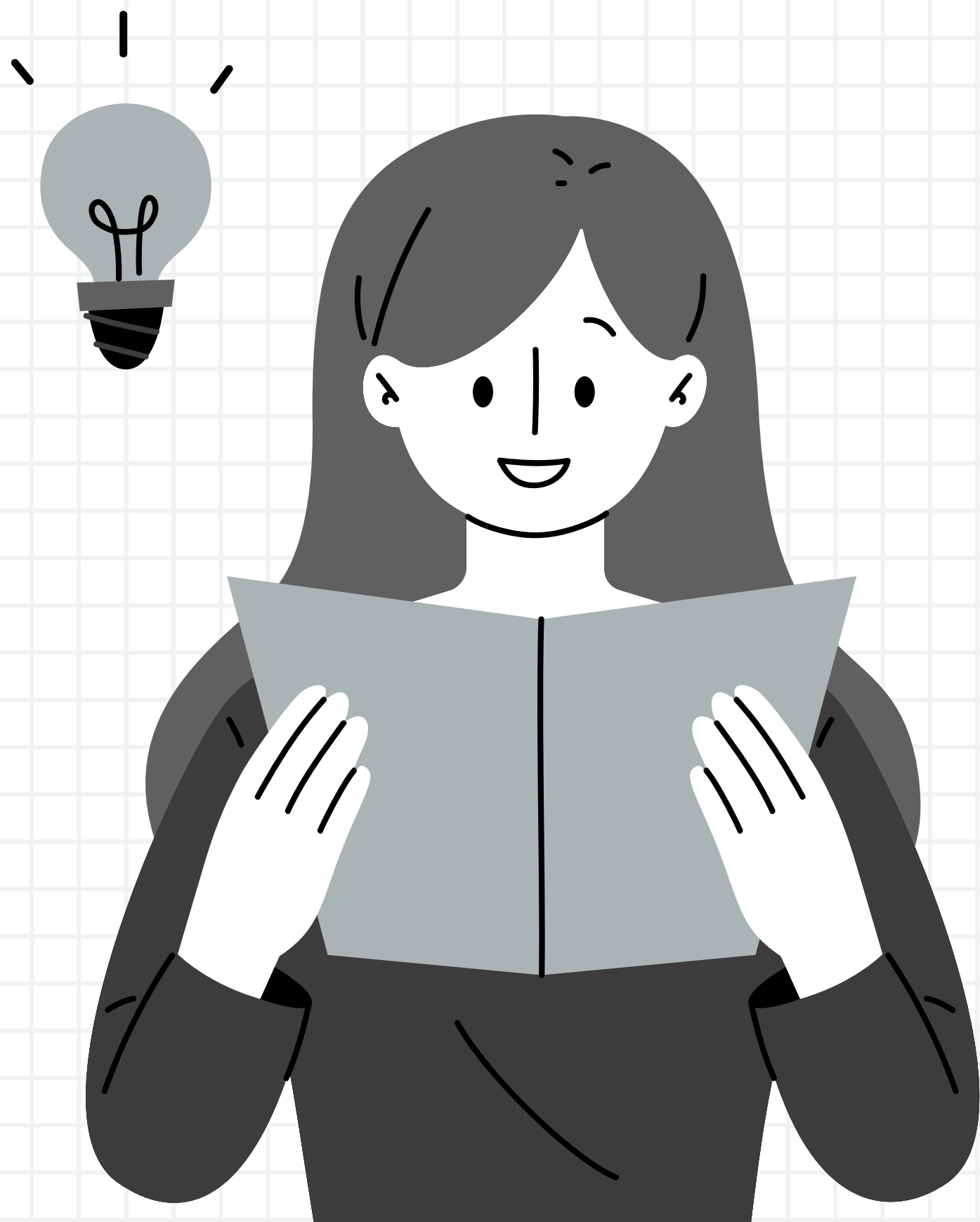
$$SHP_T(X)(\omega) = X(\omega) + K_T(\omega)$$

Backward Step:

$$SHP_t(X)(\omega) = \bigcap_{\bar{\omega} \in \text{succ}(\omega)} SHP_{t+1}(X)(\bar{\omega}) + K_t(\omega)$$

Key Idea:

- Intersect future possibilities + allow current trades
- Superhedging can be either by (both views are equivalent) →
 - trading strategies
 - consistent price systems



Thank you!